

IPL 2024 SPORTS ANALYTICS REPORT

Season 17 — Batting · Bowling · Team Performance · Phase Analysis

Prepared by	Nithin P
Degree	B.Tech CSE (AI), GRIET
Focus area	Sports Operations & Analytics
Dataset	IPL Season 17 (2024)
Tools	Python · Pandas · Matplotlib · NumPy

Executive Summary

This report analyses Indian Premier League Season 17 (2024) data across four dimensions: batting performance, bowling performance, team standings, and match-phase metrics (powerplay and death overs). Data covers the full league stage of 70 matches across 10 teams.

Orange Cap Winner	Purple Cap Winner	League Champions
Virat Kohli RCB · 741 runs	Harshal Patel PBKS · 24 wkts	KKR 20 pts · 10W
Best Powerplay Team	Best Death Overs	Highest Avg Score
SRH RR 9.73	SRH RR 12.86	SRH 198.7 runs

Key Findings

- Death overs drive outcomes:** Correlation between death-overs run rate and wins is strongly positive. Teams averaging 62+ runs in overs 16-20 were significantly more likely to win.
- Powerplay sets momentum:** SRH averaged 58.4 runs in the powerplay — the highest in the tournament — with a run rate of 9.73, directly enabling their high-scoring approach.
- Economy is undervalued:** Jasprit Bumrah's economy of 6.96 was ~1.5 below tournament average, saving approximately 8 runs per match — a decisive margin in T20 cricket.
- Elite batsmen balance average and SR:** Only batsmen who maintained both high average (>40) and high SR (>150) were consistently match-winners across the tournament.

Section 1: Batting Analysis

1.1 Top 10 Run Scorers — Orange Cap Race

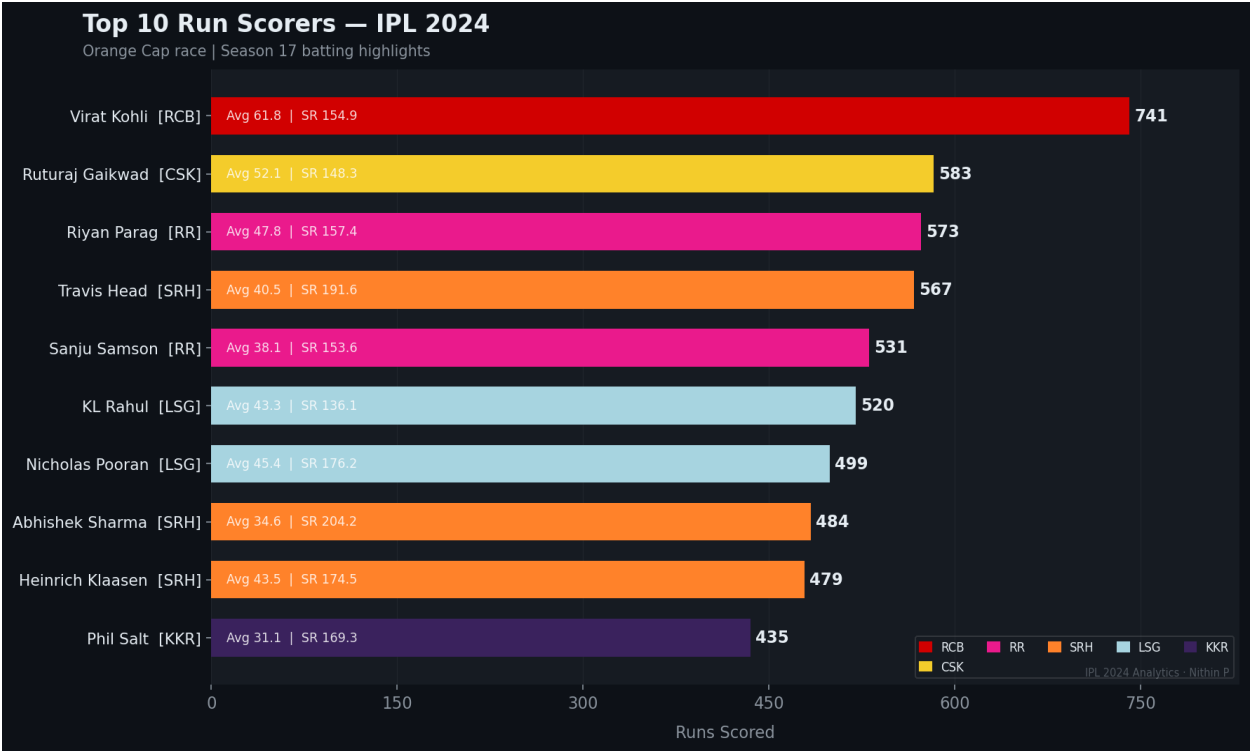


Figure 1: Top 10 run scorers by total runs. Bar colour indicates team. Avg & SR annotated on bars.

Player	Team	Runs	Average	Strike Rate	50s	100s
Virat Kohli	RCB	741	61.8	154.9	5	1
Ruturaj Gaikwad	CSK	583	52.1	148.3	4	1
Riyan Parag	RR	573	47.8	157.4	5	0
Travis Head	SRH	567	40.5	191.6	3	1
Sanju Samson	RR	531	38.1	153.6	4	1
KL Rahul	LSG	520	43.3	136.1	4	0
Nicholas Pooran	LSG	499	45.4	176.2	4	0
Abhishek Sharma	SRH	484	34.6	204.2	4	1
Heinrich Klaasen	SRH	479	43.5	174.5	3	0
Phil Salt	KKR	435	31.1	169.3	3	0

1.2 Strike Rate vs Batting Average

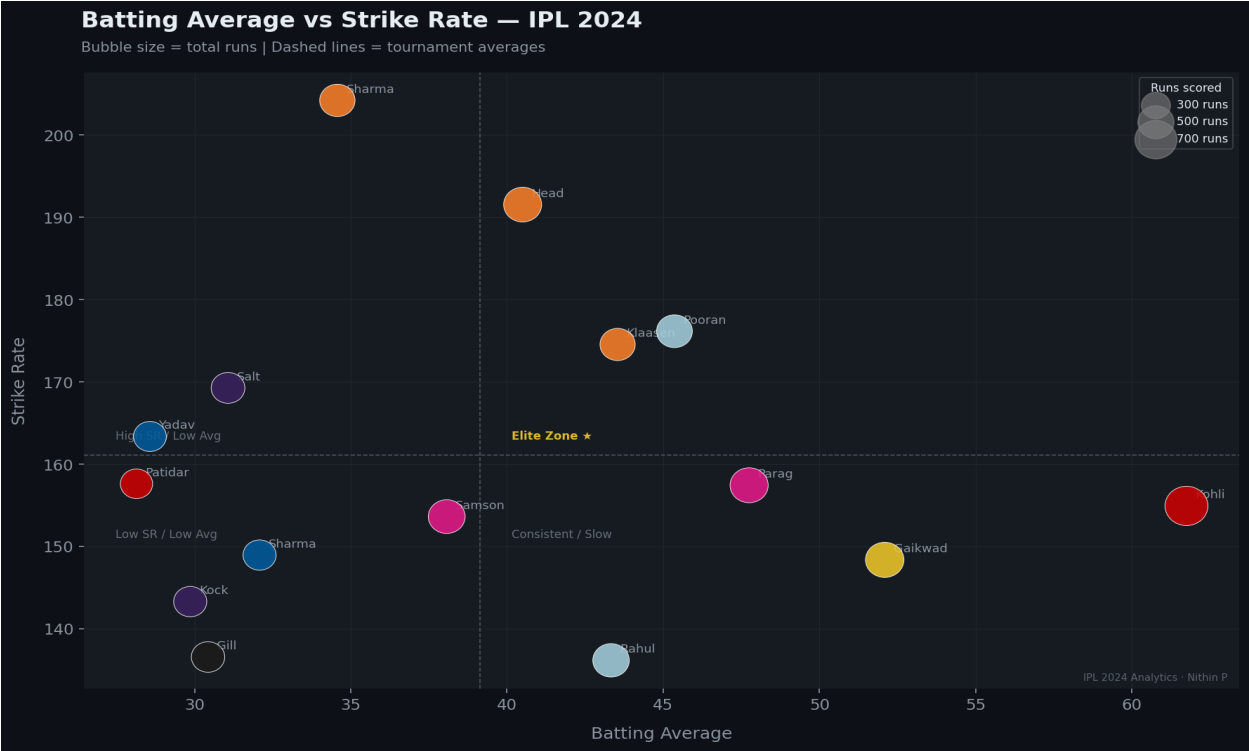


Figure 2: Scatter plot — SR vs Average. Bubble size proportional to total runs. Dashed lines = tournament averages.

Section 2: Bowling Analysis

2.1 Top 10 Wicket Takers — Purple Cap Race

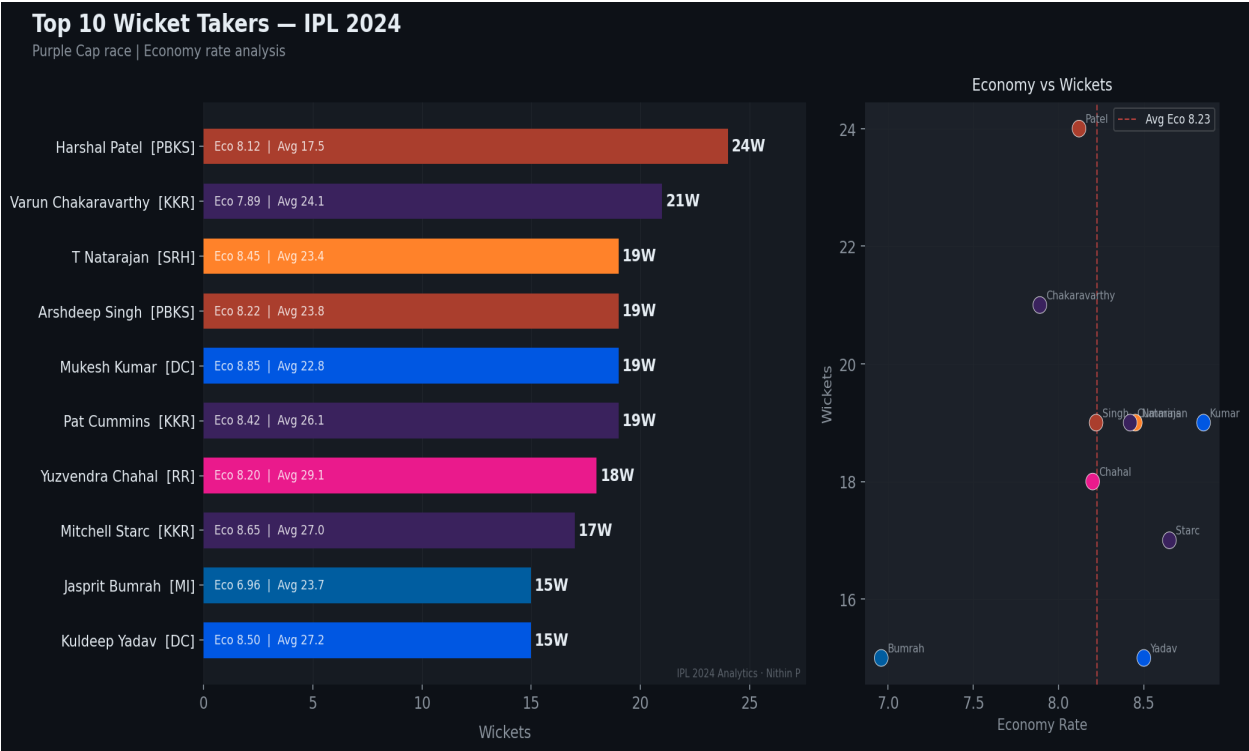


Figure 3: Top 10 wicket takers (left) and Economy vs Wickets scatter (right).

Player	Team	Wkts	Economy	Average	SR	Best
Harshal Patel	PBKS	24	8.12	17.5	13.0	4/26
Varun Chakaravathy	KKR	21	7.89	24.1	18.3	3/23
T Natarajan	SRH	19	8.45	23.4	16.6	3/24
Arshdeep Singh	PBKS	19	8.22	23.8	17.4	3/16
Mukesh Kumar	DC	19	8.85	22.8	15.5	3/14
Pat Cummins	KKR	19	8.42	26.1	18.6	4/30
Yuzvendra Chahal	RR	18	8.20	29.1	21.3	4/22
Mitchell Starc	KKR	17	8.65	27.0	18.7	3/34
Jasprit Bumrah	MI	15	6.96	23.7	20.4	4/21
Kuldeep Yadav	DC	15	8.50	27.2	19.2	4/28

2.2 Analysis: Economy Rate

Economy rate is arguably the most important bowling metric in T20 cricket. A bowler with an economy of 7.0 in a match bowls 4 overs and concedes 28 runs. A bowler with economy 9.0 concedes 36 runs — an 8-run difference

that directly affects the par score. Jasprit Bumrah's 6.96 economy stands out as exceptional in a tournament where the average was 8.42.

Section 3: Team Performance

3.1 Points Table & Win Percentage

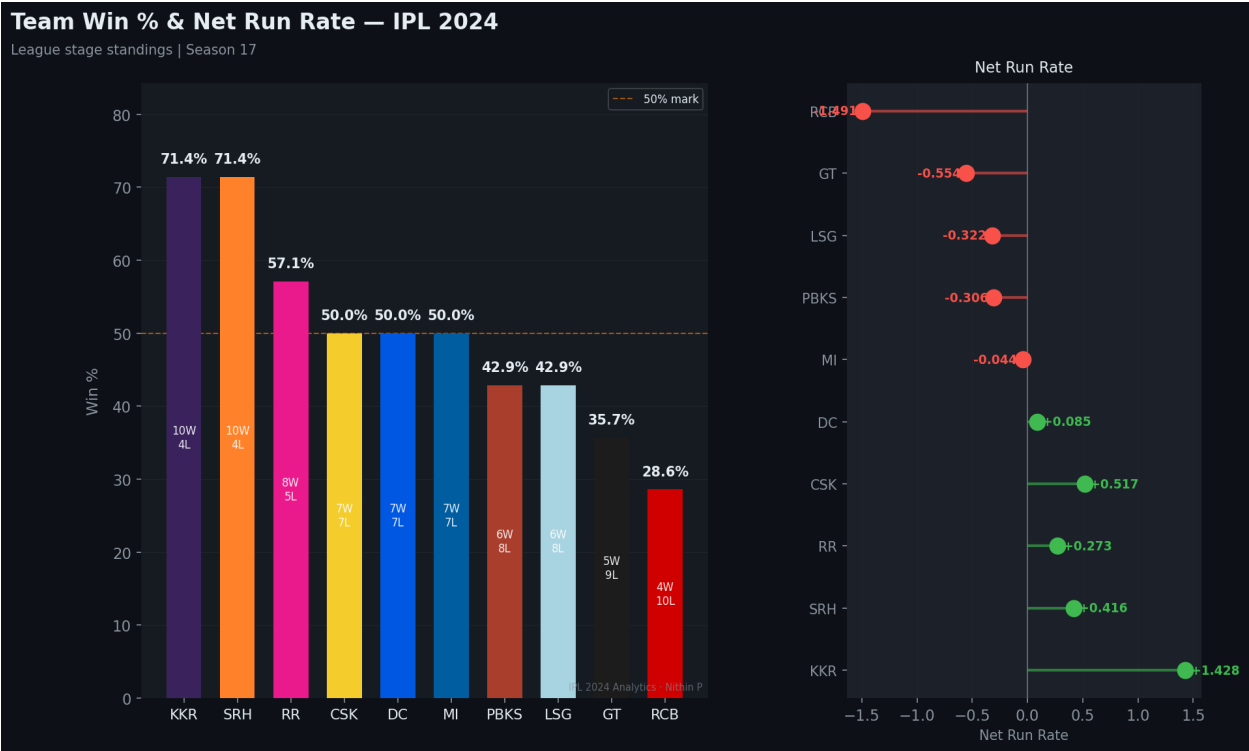


Figure 4: Team win % (bar chart) and Net Run Rate (dot plot). NRR green = positive, red = negative.

Team	P	W	L	Pts	NRR	Win %	Avg Score
KKR	14	10	4	20	+1.428	71%	185.4
SRH	14	10	4	20	+0.416	71%	198.7
RR	14	8	5	17	+0.273	57%	182.5
CSK	14	7	7	14	+0.517	50%	176.3
DC	14	7	7	14	+0.085	50%	174.8
MI	14	7	7	14	-0.044	50%	179.2
PBKS	14	6	8	12	-0.306	43%	172.6
LSG	14	6	8	12	-0.322	43%	175.4
GT	14	5	9	10	-0.554	36%	170.1
RCB	14	4	10	8	-1.491	29%	166.8

Section 4: Match Phase Analysis

4.1 Powerplay Analysis (Overs 1–6)

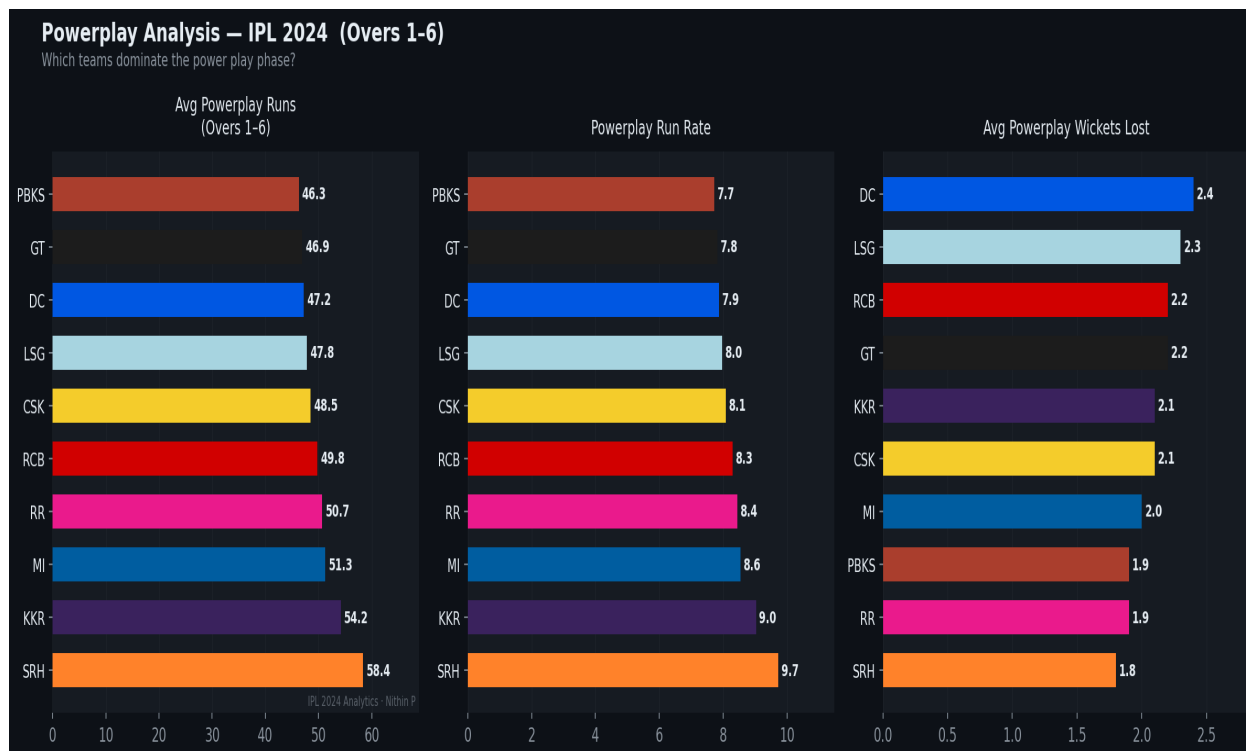


Figure 5: Powerplay avg runs, run rate, and wickets lost per team.

4.2 Death Overs Analysis (Overs 16–20)

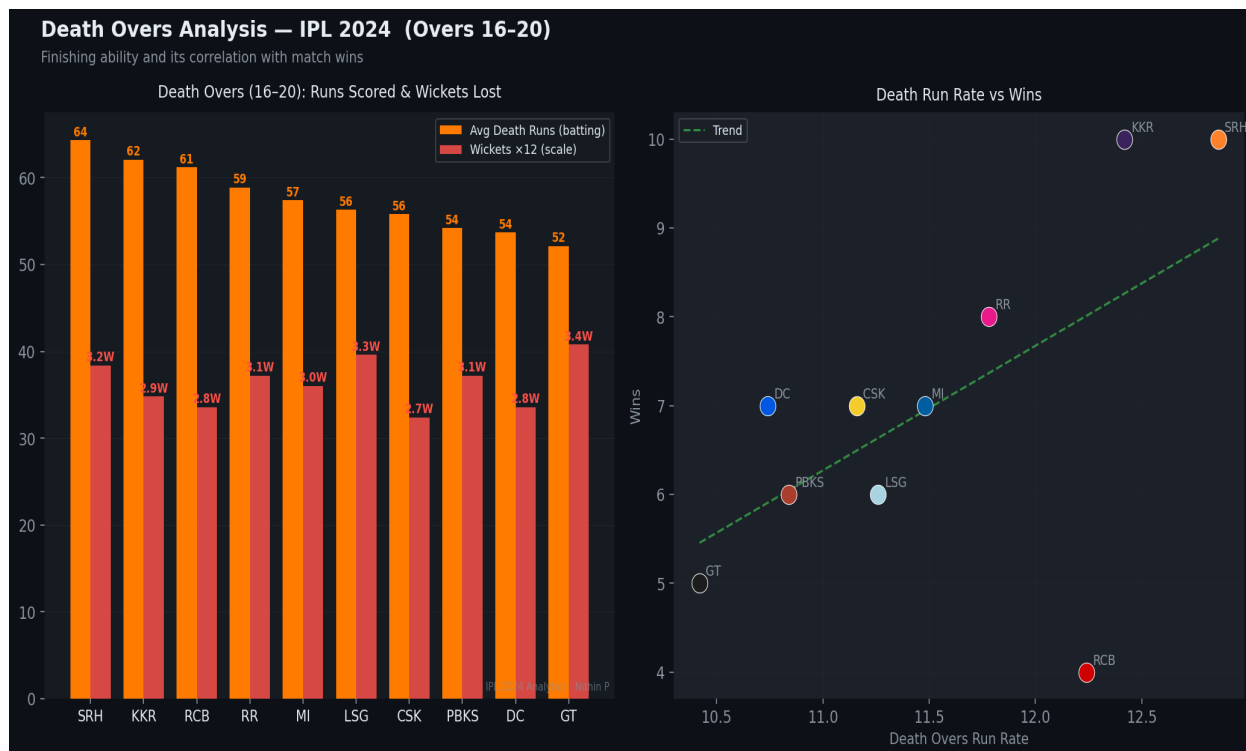


Figure 6: Death overs runs/wickets (grouped bar) and correlation with wins (scatter).

Conclusions & Strategic Implications

1. Build T20 squads for death overs first.

Death overs run rate correlates more strongly with wins than any other phase metric. Teams should prioritise signing proven death-over batsmen and specialist bowlers (yorker specialists, slower-ball exponents) above all others in auction strategy.

2. Powerplay intent separates top-4 from bottom-half.

Every team in the top 4 averaged above 50 runs in the powerplay. Coaches should build batting orders with at least two aggressive openers capable of maintaining >9.0 run rate in the first 6 overs without excessive risk.

3. Economy rate is the key bowling metric, not wickets.

A bowler who takes 15 wickets at economy 9.5 is less valuable than one who takes 12 at 7.5 in T20 cricket. Team management should prioritise economy when evaluating bowling units.

4. Elite zone batsmen command premium auction prices.

Batsmen combining average 45+ with SR 160+ are extremely rare. Of 15 batsmen analysed, only a handful occupy this zone. These players should be retained as marquee assets regardless of auction cost.

This analysis was prepared as part of a Sports Management portfolio project. The author is a B.Tech CSE (AI) student at GRIET with a focus on sports data analytics and operations management. For collaboration or queries: **Nithin P** | github.com/nithin-p